

Maximal Subcodes and Minimal Supercodes of Quasi-Cyclic Codes

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Abstract

The concatenated structure of quasi-cyclic codes allows us to determine the maximal quasi-cyclic subcodes and the minimal quasi-cyclic supercodes of a given quasi-cyclic code. We show that such subcodes and supercodes are obtained by the maximal subcodes and minimal supercodes of the constituent codes. We include the case of repeated roots. Our randomized search on the constituent codes yields many new parameters improving the best-known ones.

Key Words: Quasi-cyclic codes, subcodes, concatenated structure.

1 Preliminaries

In this section, we follow the notation in [5]. Let $\mathbb{F}_q$ be the finite field with q elements, where q is a prime power. Let m and ℓ be two positive integers. A linear code C of length $m\ell$ over $\mathbb{F}_q$ is called a *quasi-cyclic (QC) code of index ℓ* if it is invariant under the cyclic shift of codewords by ℓ positions and ℓ is the minimal number with this property. Note that, if $\ell = 1$, then C is a cyclic code. If we view codewords of C as $m \times \ell$ arrays as

$$\mathbf{c} = \begin{pmatrix} c_{0,0} & \cdots & c_{0,\ell-1} \\ \vdots & & \vdots \\ c_{m-1,0} & \cdots & c_{m-1,\ell-1} \end{pmatrix}, \quad (1.1)$$

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then being invariant under the cyclic shift by ℓ units corresponds to being closed under the row shift, where each row is moved downward one row and the bottom row moved to the top.

We consider the principal ideal $I = \langle x^m - 1 \rangle$ of $\mathbb{F}_q[x]$ and define the quotient ring $R := \mathbb{F}_q[x]/I$. If T represents the shift-by-1 operator on $\mathbb{F}_q^{m\ell}$, then its action on $\mathbf{v} \in \mathbb{F}_q^{m\ell}$ will be denoted by $T \cdot \mathbf{v}$. Hence, $\mathbb{F}_q^{m\ell}$ has an $\mathbb{F}_q[x]$ -module structure given by the multiplication

$$\begin{aligned} \mathbb{F}_q[x] \times \mathbb{F}_q^{m\ell} &\longrightarrow \mathbb{F}_q^{m\ell} \\ (a(x), \mathbf{v}) &\mapsto a(T^\ell) \cdot \mathbf{v}. \end{aligned}$$

Note that, for $a(x) = x^m - 1$, we have $a(T^\ell) \cdot \mathbf{v} = (T^{m\ell}) \cdot \mathbf{v} - \mathbf{v} = 0$. Hence, the ideal I annihilates $\mathbb{F}_q^{m\ell}$ and we can view $\mathbb{F}_q^{m\ell}$ as an R -module. Therefore, a QC code $C \subseteq \mathbb{F}_q^{m\ell}$ of index ℓ is an R -submodule of $\mathbb{F}_q^{m\ell}$.

To an element $\mathbf{c} \in \mathbb{F}_q^{m \times \ell} \simeq \mathbb{F}_q^{m\ell}$ as in (1.1), we associate an element of R^ℓ as

$$\mathbf{c}(x) := (c_0(x), c_1(x), \dots, c_{\ell-1}(x)) \in R^\ell,$$

where, for each $0 \leq j \leq \ell - 1$,

$$c_j(x) := c_{0,j} + c_{1,j}x + c_{2,j}x^2 + \dots + c_{m-1,j}x^{m-1} \in R.$$

Therefore, the following map is an R -module isomorphism:

$$\begin{aligned} \phi : \quad & \mathbb{F}_q^{m\ell} \longrightarrow R^\ell \\ \mathbf{c} = \begin{pmatrix} c_{0,0} & \dots & c_{0,\ell-1} \\ \vdots & & \vdots \\ c_{m-1,0} & \dots & c_{m-1,\ell-1} \end{pmatrix} & \longmapsto \mathbf{c}(x). \end{aligned} \quad (1.2)$$

Observe that the T^ℓ -shift on $\mathbb{F}_q^{m\ell}$ corresponds to the componentwise multiplication by x in R^ℓ . Therefore, a q -ary QC code C of length $m\ell$ and index ℓ can be considered as an R -submodule in R^ℓ .

We now describe the decomposition of a QC code over $\mathbb{F}_q$ into shorter codes over extension fields of $\mathbb{F}_q$. We refer the reader to [6, 10] for details. From this point forward, we assume that $\gcd(m, q) = 1$. Let the polynomial $x^m - 1$ factor into irreducible polynomials in $\mathbb{F}_q[x]$ as

$$x^m - 1 = f_1(x)f_2(x)\dots f_s(x). \quad (1.3)$$

Since m is relatively prime to q , there are no repeating factors in (1.3). By the Chinese Remainder Theorem (CRT), we have the following ring isomorphism:

$$R \cong \bigoplus_{i=1}^s \mathbb{F}_q[x]/\langle f_i(x) \rangle. \quad (1.4)$$

As each $f_i(x)$ divides $x^m - 1$, its roots are powers of some fixed primitive m -th root of unity, say ξ , in an extension field of $\mathbb{F}_q$. For each $i = 1, 2, \dots, s$, let u_i be the smallest nonnegative integer such that $f_i(\xi^{u_i}) = 0$. Since each $f_i(x)$ is irreducible, the direct summands in (1.4) are isomorphic to field extensions of $\mathbb{F}_q$. If $\mathbb{E}_i := \mathbb{F}_q[x]/\langle f_i(x) \rangle$, for $1 \leq i \leq s$, then we have

$$\begin{aligned} R &\cong \mathbb{E}_1 \oplus \dots \oplus \mathbb{E}_s \\ a(x) &\mapsto (a(\xi^{u_1}), \dots, a(\xi^{u_s})). \end{aligned} \quad (1.5)$$

This implies that

$$R^\ell \cong \mathbb{E}_1^\ell \oplus \dots \oplus \mathbb{E}_s^\ell.$$

Hence, a QC code $C \subseteq R^\ell$ can be viewed as an $(\mathbb{E}_1 \oplus \dots \oplus \mathbb{E}_s)$ -submodule of $\mathbb{E}_1^\ell \oplus \dots \oplus \mathbb{E}_s^\ell$ and decomposes as

$$C \cong C_1 \oplus \dots \oplus C_s, \quad (1.6)$$

where C_i is a linear code of length ℓ over $\mathbb{E}_i$, for each $i = 1, 2, \dots, s$. These length ℓ linear codes over various extensions of $\mathbb{F}_q$ are called the *constituents* of C . Let $C \subseteq R^\ell$ be generated as an R -module by

$$\{(a_{1,0}(x), \dots, a_{1,\ell-1}(x)), \dots, (a_{r,0}(x), \dots, a_{r,\ell-1}(x))\} \subset R^\ell.$$

Then, for $1 \leq i \leq s$, we have

$$C_i = \text{Span}_{\mathbb{E}_i} \{(a_{b,0}(\xi^{u_i}), \dots, a_{b,\ell-1}(\xi^{u_i})) : 1 \leq b \leq r\}. \quad (1.7)$$

We note that each extension field $\mathbb{E}_i$ above is isomorphic to a minimal cyclic code of length m over $\mathbb{F}_q$, namely, the cyclic code whose check polynomial is $f_i(x)$. If we denote by θ_i the generating primitive idempotent for the minimal cyclic code in consideration, then the isomorphism is given by the maps

$$\begin{aligned} \varphi_i : \langle \theta_i \rangle &\longrightarrow \mathbb{E}_i & \text{and} & & \psi_i : \mathbb{E}_i &\longrightarrow \langle \theta_i \rangle \\ a(x) &\longmapsto a(\xi^{u_i}) & & & \delta &\longmapsto \sum_{k=0}^{m-1} a_k x^k \end{aligned} \quad (1.8)$$

where

$$a_k = \frac{1}{m} \text{Tr}_{\mathbb{E}_i/\mathbb{F}_q}(\delta \xi^{-ku_i}).$$

Here, $\text{Tr}_{\mathbb{E}_i/\mathbb{F}_q}$ denotes the trace map from $\mathbb{E}_i$ onto $\mathbb{F}_q$. If $[\mathbb{E}_i : \mathbb{F}_q] = e_i$, then $\text{Tr}_{\mathbb{E}_i/\mathbb{F}_q}(x) = x + x^q + \dots + x^{q^{e_i-1}}$. If C_i is a length ℓ linear code over $\mathbb{E}_i$, we denote its concatenation with $\langle \theta_i \rangle$ by $\langle \theta_i \rangle \square C_i$ and the concatenation is carried out by the map ψ_i , extended to $\mathbb{E}_i^\ell$. In other words, ψ_i is applied to each symbol of the codeword in C_i to produce an element of $\langle \theta_i \rangle^\ell$.

Jensen gave the following concatenated description for QC codes.

Theorem 1.1 ([7]).

- i. Let C be an R -submodule of R^ℓ (i.e., a QC code). Then for some subset $\mathcal{I}$ of $\{1, \dots, s\}$, there exist linear codes C_i of length ℓ over $\mathbb{E}_i$, which can be explicitly described, such that

$$C = \bigoplus_{i \in \mathcal{I}} \langle \theta_i \rangle \square C_i. \quad (1.9)$$

- ii. Conversely, let C_i be a linear code in $\mathbb{E}_i^\ell$, for each $i \in \mathcal{I} \subseteq \{1, \dots, s\}$. Then,

$$C = \bigoplus_{i \in \mathcal{I}} \langle \theta_i \rangle \square C_i$$

is a q -ary QC code of length $m\ell$ and index ℓ .

It was shown in [6, Theorem 4.1] that, for a given QC code C , its constituents C_i in (1.6) and the outer codes C_i in the concatenated structure of Theorem 1.1 are equal to each other.

2 Subcodes and Supercodes of Quasi-Cyclic Codes

Given a QC code C of length $m\ell$ and index ℓ with the concatenated structure given in Theorem 1.1 above, let C' be another QC code of the same length and index such that

$$C' = \bigoplus_{i \in \mathcal{I}'} \langle \theta_i \rangle \square C'_i, \quad (2.1)$$

where $\mathcal{I}' \subseteq \{1, \dots, s\}$. Then, the following is easy to observe.

Proposition 2.1. *Given two QC codes C and C' of length $m\ell$ and index ℓ with the concatenated structures as in (1.9) and (2.1), then C' is a QC-subcode of C if and only if $\mathcal{I}' \subseteq \mathcal{I}$ and $C'_i \subseteq C_i$ for all $i \in \mathcal{I}'$. Conversely, C' is a QC-supercode of C if and only if $\mathcal{I} \subseteq \mathcal{I}'$ and $C_i \subseteq C'_i$ for all $i \in \mathcal{I}$.*

We are now ready to define maximal subcodes and minimal supercodes.

Definition 2.2. Assume that C and C' are two QC codes of length $m\ell$ and index ℓ with the concatenated structures as in (1.9) and (2.1). Allowing $\dim(C_i) = 0$, we can without loss of generality assume $\mathcal{I} = \mathcal{I}'$. Then C' is a maximal QC-subcode (minimal QC-supercode) of C if $C'_i \subsetneq C_i$ and $\dim(C'_i) = \dim(C_i) - 1$ ($C_i \subsetneq C'_i$ and $\dim(C'_i) = \dim(C_i) + 1$), for some $i \in \mathcal{I}$, and $C_j = C'_j$, for all $j \in \mathcal{I}, j \neq i$.

Assuming the notation in Section 1 above, we set $e_i := [\mathbb{E}_i : \mathbb{F}_q] = \deg(f_i(x)) = \dim(\langle \theta_i \rangle)$. Observe that Theorem 1.1 and (1.8) imply

$$m = \sum_{i=1}^s e_i \quad \text{and} \quad \dim(C) = \sum_{i \in \mathcal{I}} e_i \cdot \dim(C_i). \quad (2.2)$$

Comparing the dimension of C with the dimension of its maximal QC-subcode or minimal QC-supercode C' , we obtain the following.

Proposition 2.3. *Given two QC codes C and C' of length $m\ell$ and index ℓ such that C' is a maximal QC-subcode (minimal QC-supercode) of C , we have $\dim(C') = \dim(C) - e_i$ ($\dim(C') = \dim(C) + e_i$), for some $i \in \mathcal{I}$.*

Note that, for all $i \in \{1, \dots, s\}$, we always have $1 \leq e_i \leq m - 1$ and the least possible value $e_i = 1$ is attained by the factor $x - 1$ of $x^m - 1$.

3 The Case of Repeated Roots

Now assume that C is quasi-cyclic code of length $m\ell$ and index ℓ over $\mathbb{F}_q$ with $\gcd(q, m) = p^a$, $a > 0$. Then $m = p^a m'$ with $\gcd(m', q) = 1$. The polynomial $x^m - 1$ has the factorization

$$x^m - 1 = (x^{m'} - 1)^{p^a}. \quad (3.1)$$

Similar as in (1.3), we consider the factorization

$$x^{m'} - 1 = f_1(x)f_2(x) \dots f_s(x) \quad (3.2)$$

into irreducible factors. By the Chinese Remainder Theorem (CRT), we have the following ring isomorphism:

$$\mathbb{F}_q[x]/\langle x^m - 1 \rangle = \mathbb{F}_q[x]/\langle (x^{m'} - 1)^{p^a} \rangle \cong \bigoplus_{i=1}^s \mathbb{F}_q[x]/\langle f_i(x)^{p^a} \rangle. \quad (3.3)$$

As above, we define $\mathbb{E}_i = \mathbb{F}_q[x]/\langle f_i(x) \rangle$. From eq. (6) in [9], we obtain an isomorphism between $\mathbb{F}_q[x]/\langle f_i(x)^{p^a} \rangle$ and $S_i := \mathbb{E}_i[v]/\langle v^{p^a} \rangle$ given by

$$\Phi_i: x \mapsto \xi^{u_i}(1 - v), \quad (3.4)$$

where ξ is a primitive m' -th root of unity and $f_i(\xi^{u_i}) = 0$. The map Φ_i extends to the full ring $\mathbb{F}_q[x]/\langle x^m - 1 \rangle$.

Similar to (1.6), a quasi-cyclic code C of length $m\ell$ and index ℓ over $\mathbb{F}_q$ with $\gcd(q, m) = p^a$ decomposes as

$$C \cong C_1 \oplus \dots \oplus C_s, \quad (3.5)$$

where C_i is a linear code of length ℓ over $\mathbb{E}_i[v]/\langle v^{p^a} \rangle$, for each $i = 1, 2, \dots, s$.

Theorem 6.2 in [9] provides a somewhat involved trace description similar to the map ψ_i in (1.8). This allows to embed each of the summands C_i in (3.5) into the code C . Here we present an alternative approach, directly inverting the map Φ_i .

Lemma 3.1. *The inverse of the map Φ_i in (3.4) is given by Ψ_i which maps*

$$v \mapsto 1 - x^{1-p^a\mu} \quad (3.6)$$

$$\text{and} \quad \xi^{u_i} \mapsto x^{p^a\mu}, \quad (3.7)$$

where is such that $\xi^{u_i p^a\mu} = \xi^{u_i}$, i.e., μ is the inverse of p^a modulo m' .

Proof. First recall that ξ is an m' -th root of unity, which implies that we can choose μ as specified. The preimage of v is given in [9] after eq. (6). There it is also noted that the exponent $1 - p^a\mu$ can be taken modulo m , and hence one obtains a polynomial. Applying this partial map to the right hand side of (3.4), we obtain $\xi^{u_i} x^{1-p^a\mu}$. Since we want this to be equal to x , we have to map ξ^{u_i} , the generator of $\mathbb{E}_i = \mathbb{F}_q(\xi^{u_i})$, to $x^{p^a\mu}$. $\square$

Applying the map Ψ_i componentwise to the codes C_i in (3.5), we obtain codes over $\mathbb{F}_q[x]/\langle f_i(x)^{p^a} \rangle$. In order to invert the CRT decomposition in (3.3), the image of Ψ_i is multiplied by $g_i(x)h_i(x)$ where $g_i(x) := \prod_{i \neq j} f_j(x)^{p^a}$ and $h_i(x)$ is the inverse of $g_i(x)$ modulo $f_i(x)^{p^a}$.

It remains to study the structure of the lattice of codes over the chain ring $S_i := \mathbb{E}_i[v]/\langle v^{p^a} \rangle$. As $\mathbb{E}_i$ -vector space, we can identify S_i with $\mathbb{E}_i^{p^a}$ using

$$b(v) = \sum_{j=0}^{p^a-1} b_j v^j \mapsto (b_0, b_1, \dots, b_{p^a-1}). \quad (3.8)$$

This can be extended to an $\mathbb{E}_i$ -isomorphism between S_i^ℓ and $\mathbb{E}_i^{\ell p^a}$, where we group the coefficients according to the degree of v .

Expanding an S_i -linear code C of length ℓ , we obtain a code $\tilde{C}$ over $\mathbb{E}_i$ of length ℓp^a . Multiplication by v over S_i corresponds to shifting the expanded vector by ℓ positions to the right without wrap-around, i.e., leading positions are filled with zeros. A generator matrix for the code over $\mathbb{E}_i$ has the form

$$\begin{pmatrix} G_{1,1} & G_{1,2} & G_{1,3} & \cdots & G_{1,p^a} \\ 0 & G_{1,1} & G_{1,2} & \cdots & G_{1,p^a-1} \\ 0 & G_{2,2} & G_{2,3} & \cdots & G_{2,p^a} \\ \hline 0 & 0 & G_{1,1} & \cdots & G_{1,p^a-2} \\ 0 & 0 & G_{2,2} & \cdots & G_{2,p^a-1} \\ 0 & 0 & G_{3,3} & \cdots & G_{3,p^a} \\ \hline \vdots & \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & 0 & \cdots & G_{1,1} \\ 0 & 0 & 0 & \cdots & G_{2,2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & G_{p^a,p^a} \end{pmatrix}. \quad (3.9)$$

Each matrix $G_{j,k}$ has ℓ columns. The matrices $G_{j,j}$ for $j = 1, \dots, p^a$ have full rank. Every block between horizontal lines is repeated in the next block

shifted by ℓ positions to the right, filling the leading positions with zeros. We denote this shift operation, which corresponds to the multiplication with v , by shift_ℓ . New generators can be added to every block.

Proposition 3.2. *Let $C \subset S_i^\ell$ be a linear code over $S_i := \mathbb{E}_i[v]/\langle v^{p^a} \rangle$, i. e., an S_i -module. For a vector $\mathbf{y} \in S_i^\ell$ with $\mathbf{y} \notin C$, the code $C' = \langle C, \mathbf{y} \rangle_{S_i}$ is a minimal supercode of C if and only if $v\mathbf{y} \in C$.*

Proof. Consider the code $C'' = \langle C, v\mathbf{y} \rangle_{S_i}$. Clearly, $C \subseteq C'' \subseteq C'$, since $v\mathbf{y} \in C'$ as C' is an S_i -module. Moreover, $\mathbf{y} \notin C''$. If $v\mathbf{y} \notin C$, both inclusions are strict, i. e., $C \subsetneq C'' \subsetneq C'$, and hence C' is not a minimal supercode of C . When $v\mathbf{y} \in C$, then $\dim_{\mathbb{E}_i}(C') = 1 + \dim_{\mathbb{E}_i}(C)$, i. e., C' is a minimal supercode. $\square$

In order to find all minimal supercodes of an S_i -linear code C of length ℓ , we consider its representation as a code $\tilde{C}$ of length ℓp^a over $\mathbb{E}_i$. We compute an $\mathbb{E}_i$ -vector space complement $\tilde{V}$ of $\tilde{C}$ in $\mathbb{E}_i^{\ell p^a}$, i. e., $\tilde{C} \oplus \tilde{V} = \mathbb{E}_i^{\ell p^a}$. We take only those non-zero vectors $\tilde{\mathbf{y}} \in \tilde{V}$ for which the shift by ℓ positions is contained $\tilde{C}$. A basis for those vectors can be computed from $\tilde{C} \cap \text{shift}_\ell(\tilde{V})$.

Corollary 3.3. *Let $C \subset S_i^\ell$ be a linear code over $S_i := \mathbb{E}_i[v]/\langle v^{p^a} \rangle$, i. e., an S_i -module. Assume that $C = C' \oplus \langle \mathbf{y} \rangle$. The code C' is a maximal subcode of C if and only if $v\mathbf{y} \in C'$ and $\mathbf{y} \notin \text{shift}_\ell(C)$.*

In order to find all maximal subcodes of an S_i -linear code C of length ℓ , using Corollary 3.3 requires to compute the decomposition $C = C' \oplus \langle \mathbf{y} \rangle$ for every vector $\mathbf{y} \in C \setminus \text{shift}_\ell(C)$. Instead, we can consider the dual of minimal supercodes of the dual code $C^\perp$.

Example 3.4. We illustrate the construction of minimal supercodes starting with a distance-optimal quasi-cyclic ternary code $C = [24, 9, 10]_3$ with index $\ell = 2$ and co-index $m = 12 = 4 \times 3^1$. This is a one-generator code generated by

$$(g_1, g_2) = (x^3 + x^2 + x + 1, x^{11} + x^{10} + x^9 - x^8 - x^7 + x^5 - x^3 + x^2). \quad (3.10)$$

The polynomial $x^{12} - 1$ factors as

$$x^{12} - 1 = (x - 1)^3(x + 1)^3(x^2 + 1)^3 = f_1(x)f_2(x)f_3(x).$$

Reducing the generators in (3.10) modulo $f_i(x)$ we obtain

$$\begin{aligned} &(x^2 + x, x^2 + x), \\ &(x^2 + x - 1, -x^2), \\ &(x^3 + x^2 + x + 1, -x^4 + x^3 - x^2 + x). \end{aligned}$$

With respect to the maps Φ_i we obtain the following generators of the constituent codes C_i of length 2 over $S_i = \mathbb{E}_i/\langle v^3 \rangle$

$$\begin{aligned} &(v^2 - v, v^2 - v), \\ &(v^2 + 1, -v^2 - v - 1), \\ &(-v^2 + \omega^3 v, v^2 + \omega^3 v), \end{aligned}$$

where $\omega \in \mathbb{F}_9$ is a root of $x^2 - x - 1$. The fields are $\mathbb{E}_1 = \mathbb{E}_2 = \mathbb{F}_3$ and $\mathbb{E}_3 = \mathbb{F}_9$. Expanding these codes over S_i as codes of length 6 over $\mathbb{E}_i$ we obtain the generator matrices

$$\begin{aligned} \tilde{G}_1 &= \begin{pmatrix} 0 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}, \\ \tilde{G}_2 &= \begin{pmatrix} 1 & -1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}, \\ \tilde{G}_3 &= \begin{pmatrix} 0 & 0 & \omega^3 & \omega^3 & -1 & 1 \\ 0 & 0 & 0 & 0 & \omega^3 & \omega^3 \end{pmatrix}. \end{aligned}$$

Generator matrices of minimal supercodes as in the proof of Proposition 3.2 have the following form:

$$\begin{aligned} \tilde{G}'_1 &= \left(\begin{array}{cccccc} 0 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right), \quad \tilde{G}''_1 = \left(\begin{array}{cccccc} -1 & -1 & 1 & 1 & y_1 & y_2 \\ 0 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{array} \right), \\ \tilde{G}'_2 &= \left(\begin{array}{cccccc} 1 & -1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right), \\ \tilde{G}'_3 &= \left(\begin{array}{cccccc} 0 & 0 & \omega^3 & \omega^3 & -1 & 1 \\ 0 & 0 & 0 & 0 & \omega^3 & \omega^3 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right), \quad \tilde{G}''_3 = \left(\begin{array}{cccccc} \omega^3 & \omega^3 & -1 & 1 & z_1 & z_2 \\ 0 & 0 & \omega^3 & \omega^3 & -1 & 1 \\ 0 & 0 & 0 & 0 & \omega^3 & \omega^3 \end{array} \right), \end{aligned} \tag{3.11}$$

with $y_1, y_2 \in \mathbb{F}_3$ and $z_1, z_2 \in \mathbb{F}_9$. The horizontal line separates the new generator from the original matrix $\tilde{G}_i$. Note that using the last row of $\tilde{G}''_1$ and $\tilde{G}''_3$, respectively, one can set $y_1 = 0$ and $z_1 = 0$. Hence there are in total $(1 + 3) + 1 + (1 + 9) = 15$ different minimal supercodes. The resulting quasi-cyclic minimal supercodes are mutually non-isomorphic. There are 12 one-generator codes with parameters $[24, 10, 8]_3$, $[24, 11, 6]_3$, $[24, 11, 7]_3$, and $[24, 11, 8]_3$. The remaining 3 codes are two-generator codes with parameters $[24, 10, 6]_3$ and $[24, 11, 6]_3$. These codes corresponds to the matrices $\tilde{G}'_i$ in (3.11).

4 New Codes

4.1 New Quasi-Cyclic Codes

In this subsection, we present some new quasi-cyclic codes with improved minimum distance obtained by the methods described in Section 2 above. Namely, a search is carried out on subcodes and supercodes of a chosen constituent of some randomly generated QC code by using the computer algebra system MAGMA [2] and we verified the parameters of the new codes.

So far, we have found 70 new codes over $\mathbb{F}_3$, 71 codes over $\mathbb{F}_4$, 6 codes over $\mathbb{F}_5$, 6 codes over $\mathbb{F}_7$, 12 codes over $\mathbb{F}_8$, and 9 codes over $\mathbb{F}_9$.

The codes over $\mathbb{F}_4$ include quasi-cyclic codes of length 130 and index 2 with parameters $[130, 22, 63]_4$, $[130, 30, 56]_4$, $[130, 34, 52]_4$, $[130, 36, 50]_4$ (two-generator code), $[130, 37, 49]_4$, $[130, 38, 48]_4$, and $[130, 40, 46]_4$. These codes have sub- and supercodes with good parameters and will be used to construct additional new codes in the next subsections.

4.2 New Codes via Construction X and Variants

Similar as in [4], we use the lattice of quasi-cyclic codes to search for ingredients for Construction X and variations. Note that the subcodes and supercodes in [4] were derived from the polynomial description of quasi-cyclic codes, which results in a coarse grained lattice.

First we recall Construction X (see [11, Ch. 18, §7.1]):

Proposition 4.1 (Construction X). *Given a code $C_1 = [n, k_1, d_1]_q$, a subcode $C_2 \subset C_1$ with parameters $C_2 = [n, k_2, d_2]_q$, and an auxiliary code $C_3 = [n', k_1 - k_2, d_3]_q$, there is a code C with parameters $C = [n + n', k_1, d]_q$ with $d \geq \min(d_1 + d_3, d_2)$.*

From nested quasi-cyclic codes of length 130 and index 2 over $\mathbb{F}_4$ we obtain the new codes shown in Table 1.

Alltop [1] proposed the following variation of Construction X:

Proposition 4.2 (Construction XX). *Given $C_1 = [n, k_1, d_1]_q$ with subcodes C_2 and C_4 with parameters $C_2 = [n, k_2, d_2]_q$ and $C_3 = [n, k_3, d_3]_q$, respectively, let $C_4 = C_2 \cap C_3$ with parameters $C_4 = [n, k_4, d_4]_q$. Using auxiliary codes $D_{12} = [n', k_1 - k_2, d']_q$ and $D_{13} = [n'', k_1 - k_3, d'']_q$, there is a code C with parameters $C = [n + n' + n'', k_1, d]_q$ with $d \geq \min\{d_1 + d' + d'', d_2 + d'', d_3 + d', d_4\}$.*

Again using nested quasi-cyclic codes of length 130 and index 2 over $\mathbb{F}_4$, we obtained the new codes shown in Table 2.

Remark 4.3. The methods presented here can also be applied to quasi-twisted codes [3]. More precisely, using the concatenated description of quasi-twisted codes given in [8], we determine their maximal subcodes and

minimal supercodes in terms of their constituents. Then, our findings can be used as ingredients for Construction X and its variants.

Table 1: New codes obtained via Construction X applied to quasi-cyclic codes of length 130.

C	C_1	C_2	C_3
$[132, 22, 64]_4$	$[130, 22, 63]_4$	$[130, 20, 64]_4$	$[2, 2, 1]_4$
$[132, 34, 52]_4$	$[130, 34, 51]_4$	$[130, 32, 52]_4$	$[2, 2, 1]_4$
$[133, 20, 68]_4$	$[130, 20, 66]_4$	$[130, 18, 68]_4$	$[3, 2, 2]_4$
$[133, 32, 54]_4$	$[130, 32, 52]_4$	$[130, 30, 54]_4$	$[3, 2, 2]_4$
$[134, 22, 66]_4$	$[130, 22, 63]_4$	$[130, 20, 66]_4$	$[4, 2, 3]_4$
$[134, 34, 54]_4$	$[130, 34, 51]_4$	$[130, 32, 54]_4$	$[4, 2, 3]_4$
$[134, 36, 51]_4$	$[130, 36, 48]_4$	$[130, 34, 51]_4$	$[4, 2, 3]_4$
$[135, 16, 75]_4$	$[130, 16, 71]_4$	$[130, 14, 76]_4$	$[5, 2, 4]_4$
$[137, 32, 56]_4$	$[130, 32, 54]_4$	$[130, 26, 58]_4$	$[7, 6, 2]_4$
$[137, 37, 51]_4$	$[130, 37, 49]_4$	$[130, 31, 54]_4$	$[7, 6, 2]_4$
$[137, 40, 48]_4$	$[130, 40, 46]_4$	$[130, 34, 50]_4$	$[7, 6, 2]_4$
$[138, 36, 52]_4$	$[130, 36, 48]_4$	$[130, 32, 52]_4$	$[8, 4, 4]_4$
$[140, 32, 58]_4$	$[130, 32, 54]_4$	$[130, 26, 58]_4$	$[10, 6, 4]_4$
$[140, 40, 50]_4$	$[130, 40, 46]_4$	$[130, 34, 50]_4$	$[10, 6, 4]_4$
$[141, 33, 57]_4$	$[130, 33, 52]_4$	$[130, 27, 57]_4$	$[11, 6, 5]_4$
$[141, 37, 54]_4$	$[130, 37, 49]_4$	$[130, 31, 54]_4$	$[11, 6, 5]_4$
$[145, 34, 58]_4$	$[130, 34, 52]_4$	$[130, 26, 58]_4$	$[15, 8, 6]_4$

Table 2: New codes obtained via Construction XX applied to quasi-cyclic codes of length 130.

C	C_1	C_2	C_3	C_4	D_{12}	D_{13}
$[136, 22, 67]_4$	$[130, 22, 63]_4$	$[130, 20, 64]_4$	$[130, 20, 66]_4$	$[130, 18, 68]_4$	$[2, 2, 1]_4$	$[4, 2, 3]_4$
$[136, 36, 52]_4$	$[130, 36, 48]_4$	$[130, 34, 49]_4$	$[130, 34, 51]_4$	$[130, 32, 52]_4$	$[2, 2, 1]_4$	$[4, 2, 3]_4$
$[138, 22, 68]_4$	$[130, 22, 63]_4$	$[130, 20, 64]_4$	$[130, 20, 66]_4$	$[130, 18, 68]_4$	$[3, 3, 2]_4$	$[5, 2, 4]_4$
$[143, 35, 56]_4$	$[130, 35, 50]_4$	$[130, 29, 52]_4$	$[130, 32, 54]_4$	$[130, 26, 56]_4$	$[7, 6, 2]_4$	$[6, 3, 4]_4$
$[144, 34, 58]_4$	$[130, 34, 51]_4$	$[130, 32, 54]_4$	$[130, 28, 56]_4$	$[130, 26, 58]_4$	$[4, 2, 3]_4$	$[10, 6, 4]_4$
$[146, 33, 59]_4$	$[130, 33, 51]_4$	$[130, 30, 54]_4$	$[130, 27, 56]_4$	$[130, 24, 59]_4$	$[5, 3, 3]_4$	$[12, 6, 6]_4$
$[150, 35, 60]_4$	$[130, 35, 50]_4$	$[130, 33, 52]_4$	$[130, 27, 58]_4$	$[130, 25, 60]_4$	$[3, 2, 2]_4$	$[17, 8, 8]_4$

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